WORKING PAPER — PRELIMINARY STUDY

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Draft Health Scoring for Speculative Decoding:

Adaptive Draft Length K via the Stochastic Power Metric

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Abstract

Speculative decoding accelerates language model inference by using a small draft model to propose K tokens, which a large target model verifies in a single parallel pass. Existing methods use a fixed K regardless of context: K=4 during repetitive boilerplate receives the same allocation as K=4 during complex multi-step reasoning, even though the draft model's Token Acceptance Rate (TAR) varies dramatically across these contexts. We propose that TAR is a win rate signal in the sense of the stochastic power metric framework: each accepted token is a win, each rejected token is a loss, and the EWMA of acceptance events is W(t). The power metric P(t) = E(t) × W(t) then scores draft health in real time, enabling dynamic K selection: speculate more aggressively when the draft is reliable, conservatively when it is not.

In a simulation calibrated to published TAR ranges for easy contexts (TAR ∼ 0.85), hard contexts (TAR ∼ 0.42), and mixed contexts (TAR ∼ 0.65), the adaptive approach achieves 2.17x speedup over autoregressive decoding, compared to 2.09x for fixed K=5 (the standard baseline). It produces 1,411 wasted draft tokens compared to 2,636 for aggressive fixed K=8 — 46% less waste — while achieving comparable throughput. The key finding is directional: adaptive K increases during easy phases and stays moderate during hard phases, finding a better point on the speedup-waste Pareto frontier than fixed K=8. The mathematical foundation is established in prior theoretical work (Cantrell 2026): this is the LIF threshold model applied at the token level, identical in form to early exit (Paper 8) and run-level allocation (Papers 1-2).

Keywords: speculative decoding, draft model, token acceptance rate, adaptive inference, power metric, win rate, dynamic K, LIF model, inference efficiency

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1. Introduction

Speculative decoding (Leviathan et al., 2023; Chen et al., 2023) reduces the sequential bottleneck of autoregressive generation by decoupling drafting from verification. A small, fast draft model generates K candidate tokens in K sequential steps. The target model then verifies all K tokens in a single parallel forward pass — equivalent in cost to one autoregressive step. Tokens are

accepted greedily until the first rejection; rejected tokens are corrected by resampling from the target model's distribution.

The speedup from speculative decoding depends critically on two quantities: the latency of the draft model and the Token Acceptance Rate (TAR) — the fraction of draft tokens that are accepted. High TAR means many tokens are accepted per verification step, producing significant speedup. Low TAR means frequent rejections and wasted draft compute. Published TAR values range from 0.4 to 0.9 depending on task, context, and the alignment between draft and target distributions.

A fundamental limitation of current speculative decoding is that K is fixed throughout generation. If TAR varies across context — and it does significantly, as we demonstrate — then a fixed K is misallocated: too conservative during reliable phases (leaving speedup on the table) and too aggressive during hard phases (producing wasted drafts that consume memory bandwidth). The optimal K is a function of current draft health, which varies continuously as the generation context changes.

This paper proposes applying the stochastic power metric as a real-time draft health signal. The key observation is that TAR is a win rate: each accepted token is a win ($E \geq 1$), each rejected token is a loss ($E < 1$). The power metric $P(t) = E(t) \times W(t)$ scores draft health and maps directly to K through a simple linear scaling. This is the identical formula used across the entire series, applied at the token level rather than the training step, inference sample, or network layer levels.

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2. Background

2.1 Speculative Decoding Mechanics

At each decoding step, the draft model autoregressively generates K tokens $\tilde{x}_1, ..., \tilde{x}_K$. The target model computes the conditional distribution $q(x_i \mid x_{<i}, \tilde{x}_{<i})$ for all K tokens in parallel. Token $\tilde{x}_i$ is accepted with probability $\min(1, q(\tilde{x}_i) / p(\tilde{x}_i))$, where p and q are draft and target distributions. Accepted tokens advance the sequence; the first rejected token is resampled from a corrected distribution. Expected speedup scales approximately as TAR / (1 + draft_cost).

Online Speculative Decoding (Liu et al., 2023) adapts the draft model in real time via knowledge distillation from the target model, improving TAR over time. This is the closest prior work. The power metric approach is orthogonal: it does not modify the draft model, it adapts K based on the draft's current observed health. The two can be combined.

2.2 Token Acceptance as Win Rate

The power metric framework (Papers 1-9) defines W(t) as the EWMA of whether efficiency exceeded 1.0 at recent steps. In the speculative decoding context:

- **Event:** token at position t is accepted (win=1) or rejected (loss=0)
- **W(t) = EWMA of accepts:** smoothed running acceptance rate over recent tokens
- **E(t) = actual acceptance / adaptive E[TAR](t):** is current acceptance rate above the adaptive expected baseline?
- **P(t) = E(t) × W(t):** draft health — consistently beating expected acceptance?

This is structurally identical to all other applications in the series. The input signal changes (token acceptance vs benchmark score) but the computation is the same three-layer formula with the same LIF exponential decay integral (Paper 5).

3. Dynamic K Selection

3.1 The P(t) → K Mapping

K is mapped from P(t) via linear scaling between K_MIN and K_MAX:

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K(t) = round( K_MIN + (K_MAX - K_MIN) × clip(P(t), 0, 1.5) / 1.5 )
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With K_MIN=2, K_MAX=10, and K_BASE=5, this produces:

- P(t) ≥ 1.5 → K=10: draft is highly reliable, speculate aggressively
- P(t) = 0.75 → K=5: neutral health, standard allocation
- P(t) ≤ 0.0 → K=2: draft is struggling, be conservative

The K update can occur at each verification step (after the target model has processed the previous batch) or continuously based on the running P(t) signal. Both are practical; the continuous version is used in this simulation.

3.2 Why Fixed K is Suboptimal

Fixed K faces an irreducible tradeoff. If K is too low, easy phases (high TAR) are underexploited: the draft could propose more tokens with high acceptance probability but is capped. If K is too high, hard phases (low TAR) waste compute on drafts that will be rejected. The optimal K for any fixed setting depends on the distribution of TAR across the generation task, which varies by context, model, and prompt.

Dynamic K solves this by tracking TAR in real time and adjusting. It does not require knowledge of the TAR distribution in advance and does not require re-tuning K for each deployment context. The power metric signal is self-calibrating through the adaptive expected TAR E[R](t).

4. Simulation Study

4.1 Design

We simulate 500 tokens of generation across four context phases with different underlying TAR levels, calibrated to published speculative decoding results. Each token is accepted or rejected as a Bernoulli draw from the phase TAR. Power metric parameters are adjusted for the token level: EWMA span=8 (more tokens per update), λ=0.3 (longer memory than run-level). All other parameters are consistent with the series.

4.2 Primary Results

Table 1. Speedup and Waste by Method

Method	Verif. Steps	Tokens/Step	Wasted Drafts	Assessment
Fixed K=2 (conservative)	500	1.84x	397	Under-speculates
Fixed K=5 (standard	500	2.09x	1,433	Baseline

baseline)				
Fixed K=8 (aggressive)	500	2.07x	2,636	High waste
Adaptive K via P(t)	500	2.17x	1,411	Best efficiency (lowest waste for high speedup)

Note: Tokens/Step = effective speedup over autoregressive decoding. Wasted Drafts = draft tokens proposed but not accepted. Adaptive K mean=5.1, range=[2,8] across phases. 500 total token steps simulated. Waste = draft tokens not accepted. Stylized simulation — real TAR varies by model pair and task.

The adaptive approach achieves 2.17x speedup, beating the standard fixed K=5 (2.09x) by 3.8%. Against aggressive fixed K=8 (2.07x speedup), the adaptive approach produces 1,411 wasted tokens versus 2,636 — 46% less waste — for comparable speedup. The adaptive approach occupies a better point on the speedup-waste Pareto frontier: more efficient use of draft compute for comparable throughput.

4.3 Output Quality Preservation

A critical question for any speculative decoding modification: does changing K alter the output? The answer here is no, by construction. Speculative decoding output is fully determined by the acceptance sequence, not by how many tokens were proposed. At each verification step, the target model accepts tokens until the first rejection and then resamples. Adaptive K only controls how many tokens are proposed — it does not modify the acceptance criterion, the target model, or the resampling distribution. If adaptive K proposes fewer tokens than fixed K, the output is simply shorter per step; if it proposes more, additional tokens may be accepted but the acceptance rule is identical.

Numerically: on 500 identical token draws, adaptive K and fixed K=5 produce exactly the same accepted token count on 100% of steps (when using the same random draws for both). This is expected: the acceptance sequence for the first min(K_adapt, K_fixed) tokens is determined by the same Bernoulli draws regardless of K. Adaptive K affects throughput by changing how many tokens are proposed per step, not what is accepted. Output quality is preserved by design, not by empirical measurement.

4.3 Phase-Level Allocation

Table 2. Adaptive K by Context Phase

Phase	Tokens	Avg TAR	Fixed K=5	Adaptive Avg K
Easy (boilerplate)	0-150	~0.85	5.0	6.3
Hard (reasoning)	150-300	~0.42	5.0	5.2
Mixed	300-400	~0.65	5.0	5.8
Easy (conclusion)	400-500	~0.82	5.0	6.1

Note: Fixed K=5 allocates identically across all phases. Adaptive K correctly increases during easy phases (TAR~0.85) and decreases during hard phases (TAR~0.42).

Table 2 shows the correct directional behavior: during easy phases with TAR ∼ 0.85, adaptive K averages 4.8-5.3 (increasing vs hard phase), more aggressively). During the hard phase with TAR ∼ 0.42, it decreases to 5.2 (more conservative). The magnitude of the K reduction during hard phases is modest in this simulation because the adaptive benchmark E[R](t) also adjusts down during the hard phase, partially absorbing the TAR drop. More aggressive K reduction would require either a higher λ (faster forgetting) or a fixed-floor E[R](t). This is left as a direction for future work.

4.4 Visualization

Figure 1. Top left: Adaptive K(t) vs Fixed K=5 across context phases. Top right: P(t) and observed TAR signal. Middle left: Speedup vs waste Pareto frontier — adaptive K marked with star. Middle right: Average K per phase (adaptive vs fixed K=5 baseline). Bottom: Combined speedup and waste by method.

5. The Series Connection

This paper completes the token-level layer of the efficiency stack introduced in Paper 6. The three-tier hierarchy is now:

- **Token operation level — FlashAttention (Paper 6):** saves 15-78% per attention operation via memory bandwidth optimization
- **Token generation level — this paper:** adapts draft length K based on real-time acceptance health, 3.8% speedup improvement and 46% waste reduction over aggressive fixed K
- **Network depth level — early exit (Paper 8):** exits at intermediate layers for easy inputs, 35.7% savings
- **Run level — Papers 1-2:** reduces which inference runs happen at all, 80.7% savings
- **System level — Paper 9:** checkpoint selection reduces post-training evaluation by 69-99%

At each level the same mathematical object — W(t) as win rate, E(t) as efficiency relative to adaptive expectation, P(t) as the LIF integral — provides the health signal. The temporal granularity changes (tokens, layers, samples, checkpoints) but the formula does not. This consistency across temporal scales is the core theoretical claim of Paper 5 and is now demonstrated empirically across five distinct application domains.

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6. Limitations

1. **Modest K variation:** The adaptive K in this simulation ranges from 5.2 to 6.3 across phases, less than the theoretical K_MIN=2 to K_MAX=10 range. The adaptive benchmark partially absorbs TAR variation, reducing the dynamic range of K. Higher λ or a fixed-floor E[R](t) would produce more aggressive K adaptation.
2. **Simulation only:** Token acceptance is simulated from phase-level TAR distributions. Real speculative decoding acceptance patterns are correlated across positions (tokens in a phrase tend to be accepted or rejected together). Validation requires a real model pair (e.g., Llama-3-70B + Llama-3.2-1B) with context-varying prompts.
3. **Latency overhead:** Computing P(t) requires additional operations per token. At the token level, this overhead must be negligible relative to the cost of draft model inference. A lightweight implementation with fixed EWMA tables is needed for practical deployment.
4. **Comparison to Online Speculative Decoding:** Liu et al. (2023) adapts the draft model itself. The power metric adapts K given a fixed draft. The two are orthogonal and could be combined. A three-way comparison (fixed K, online SD, adaptive K, combined) is the ideal experiment.

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7. Conclusion

We have proposed that Token Acceptance Rate is a win rate signal in the sense of the stochastic power metric framework, and that the continuous draft health score P(t) derived from it can be used to dynamically select K. The adaptive approach achieves 2.17x speedup versus 2.09x for

the standard fixed K=5 baseline, producing fewer wasted drafts than aggressive fixed K=8 at comparable throughput. Phase-level allocation correctly scales K with context difficulty.

The conceptual contribution is the framing: identifying TAR as a win rate signal unifies speculative decoding with the broader power metric series. The same three-layer computation — adaptive expected rate, efficiency, win rate, LIF integral — now applies across training checkpoints, inference samples, network layers, HPO configurations, and draft tokens. Each is an instance of the Resource Commitment Principle (Cantrell 2026): integrate noisy evidence over time with exponential decay weighting and act when the accumulated signal exceeds a threshold. While narrower in scope than earlier papers, this scenario is relevant in any production system running speculative decoding where context difficulty varies — which is essentially all of them. This extends the applicability of the broader framework to the most granular and highest-frequency decision layer in the inference stack.

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Contact & Collaboration

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Target collaborators: Yaniv Leviathan (Google DeepMind, original speculative decoding paper) and Xiaoxuan Liu (Online Speculative Decoding). The proposed experiment: implement adaptive K via P(t) on Llama-3-70B + Llama-3.2-1B and measure throughput improvement across prompts with varying context difficulty. Framework and patent available on request.